Introduction

This exercise is designed to build proficiency in SQL date functions across five progressive skill areas: extracting components from date values, converting between text and date data types, performing date arithmetic and interval calculations, filtering datasets by temporal criteria, and applying conditional logic to date-derived values. Through twenty structured questions plus a bonus challenge, learners will work with realistic business scenarios and order processing, customer retention, subscription management, and financial reporting to develop practical fluency rather than purely theoretical knowledge.

Question 1: Day Name from Order Date

```sql
-- Q1: Day Name from Order Date
CREATE OR REPLACE TABLE orders (
  order_id    INT,
  customer_id INT,
  order_date  DATE
);
INSERT INTO orders VALUES
  (1001, 101, '2026-05-01'),
  (1002, 102, '2026-05-02'),
  (1003, 103, '2026-05-03'),
  (1004, 104, '2026-05-04'),
  (1005, 105, '2026-05-05');

SELECT
  order_id,
  customer_id,
  order_date,
  DAYNAME(order_date) AS day_name
FROM orders;
```

Add parameter

Results 3 of 3 | Table

	order_id	customer_id	order_date	day_name
1	1001	101	2026-05-01	Fri
2	1002	102	2026-05-02	Sat
3	1003	103	2026-05-03	Sun
4	1004	104	2026-05-04	Mon
5	1005	105	2026-05-05	Tue

Question 2: Month Name from Signup Date

```sql
-- Q2: Month Name from Signup Date
CREATE OR REPLACE TABLE customer_signups (
  customer_id   INT,
  customer_name STRING,
  signup_date   DATE
);
INSERT INTO customer_signups VALUES
  (201, 'John',  '2026-01-15'),
  (202, 'Mary',  '2026-02-20'),
  (203, 'Peter', '2026-03-05'),
  (204, 'Sarah', '2026-04-18'),
  (205, 'Thabo', '2026-05-30');

SELECT
  customer_id,
  customer_name,
  signup_date,
  MONTHNAME(signup_date) AS signup_month_name
FROM customer_signups;
```

Add parameter

Results 3 of 3 | Table

	customer_id	customer_name	signup_date	signup_month_name
1	201	John	2026-01-15	Jan
2	202	Mary	2026-02-20	Feb
3	203	Peter	2026-03-05	Mar
4	204	Sarah	2026-04-18	Apr
5	205	Thabo	2026-05-30	May

Question 3: Month Number from Sale Date

```sql
-- Q3: Month Number from Sale Date
CREATE OR REPLACE TABLE sales (
  sale_id      STRING,
  product_name STRING,
  sale_date    DATE,
  amount       DECIMAL(10,2)
);
INSERT INTO sales VALUES
  ('S001', 'Laptop',   '2026-01-10', 12000),
  ('S002', 'Mouse',    '2026-02-15', 250),
  ('S003', 'Keyboard', '2026-03-20', 700),
  ('S004', 'Monitor',  '2026-04-25', 3500),
  ('S005', 'Desk',     '2026-05-30', 2000);

SELECT
  sale_id,
  product_name,
  sale_date,
  MONTH(sale_date) AS sale_month
FROM sales;
```

Add parameter

Results 3 of 3 | Table

	sale_id	product_name	sale_date	sale_month
1	S001	Laptop	2026-01-10	1
2	S002	Mouse	2026-02-15	2
3	S003	Keyboard	2026-03-20	3
4	S004	Monitor	2026-04-25	4
5	S005	Desk	2026-05-30	5

Question 4: Year from Transaction Date

```sql
-- Q4: Year from Transaction Date
CREATE OR REPLACE TABLE transactions (
  transaction_id   STRING,
  customer_id      INT,
  transaction_date DATE,
  amount           DECIMAL(10,2)
);
INSERT INTO transactions VALUES
  ('T001', 101, '2024-12-15', 500),
  ('T002', 102, '2025-01-20', 1200),
  ('T003', 103, '2025-06-10', 800),
  ('T004', 104, '2026-02-05', 1500),
  ('T005', 105, '2026-05-25', 2000);

SELECT
  transaction_id,
  customer_id,
  transaction_date,
  YEAR(transaction_date) AS transaction_year
FROM transactions;
```

Add parameter

Results 3 of 3 | Table

	transaction_id	customer_id	transaction_date	transaction_year
1	T001	101	2024-12-15	2024
2	T002	102	2025-01-20	2025
3	T003	103	2025-06-10	2025
4	T004	104	2026-02-05	2026
5	T005	105	2026-05-25	2026

Question 5: Day of Month from Delivery Date

```sql
-- Q5: Day of Month from Delivery Date
CREATE OR REPLACE TABLE deliveries (
  delivery_id   STRING,
  customer_id   INT,
  delivery_date DATE
);
INSERT INTO deliveries VALUES
  ('D001', 101, '2026-05-01'),
  ('D002', 102, '2026-05-08'),
  ('D003', 103, '2026-05-15'),
  ('D004', 104, '2026-05-22'),
  ('D005', 105, '2026-05-29');

SELECT
  delivery_id,
  customer_id,
  delivery_date,
  DAY(delivery_date) AS day_of_month
FROM deliveries;

-- ================================================================================
```

Add parameter

Results 3 of 3 | Table

	delivery_id	customer_id	delivery_date	day_of_month
1	D001	101	2026-05-01	1
2	D002	102	2026-05-08	8
3	D003	103	2026-05-15	15
4	D004	104	2026-05-22	22
5	D005	105	2026-05-29	29

Question 6: Add Today's Date

```sql
-- Q6: Add Today's Date
CREATE OR REPLACE TABLE employees (
  employee_id   INT,
  employee_name STRING,
  department    STRING
);
INSERT INTO employees VALUES
  (301, 'Nandi', 'Sales'),
  (302, 'Brian', 'IT'),
  (303, 'Lerato', 'Finance'),
  (304, 'Sipho', 'HR'),
  (305, 'Aisha', 'Marketing');

SELECT
  employee_id,
  employee_name,
  department,
  CURRENT_DATE() AS today_date
FROM employees;
```

Add parameter

Results 3 of 3 | Table

	employee_id	employee_name	department	today_date
1	301	Nandi	Sales	2026-08-01
2	302	Brian	IT	2026-08-01
3	303	Lerato	Finance	2026-08-01
4	304	Sipho	HR	2026-08-01
5	305	Aisha	Marketing	2026-08-01

Question 7: Convert Text to Date

```sql
-- Q7: Convert Text to Date
CREATE OR REPLACE TABLE online_orders (
  order_id        INT,
  customer_id     INT,
  order_date_text STRING
);
INSERT INTO online_orders VALUES
  (4001, 101, '2026-01-15'),
  (4002, 102, '2026-02-20'),
  (4003, 103, '2026-03-25'),
  (4004, 104, '2026-04-10'),
  (4005, 105, '2026-05-05');

SELECT
  order_id,
  customer_id,
  order_date_text,
  TO_DATE(order_date_text) AS order_date
FROM online_orders;
```

Add parameter

Results 3 of 3 | Table

	order_id	customer_id	order_date_text	order_date
1	4001	101	2026-01-15	2026-01-15
2	4002	102	2026-02-20	2026-02-20
3	4003	103	2026-03-25	2026-03-25
4	4004	104	2026-04-10	2026-04-10
5	4005	105	2026-05-05	2026-05-05

Question 8: Format Date as Text

```sql
-- Q8: Format Date as Text
CREATE OR REPLACE TABLE payment_dates (
  payment_id   STRING,
  customer_id  INT,
  payment_date DATE
);
INSERT INTO payment_dates VALUES
  ('P001', 101, '2026-01-05'),
  ('P002', 102, '2026-02-10'),
  ('P003', 103, '2026-03-15'),
  ('P004', 104, '2026-04-20'),
  ('P005', 105, '2026-05-25');

SELECT
  payment_id,
  customer_id,
  payment_date,
  DATE_FORMAT(payment_date, 'yyyy-MM-dd') AS formatted_payment_date
FROM payment_dates;
```

Add parameter

Results 3 of 3 | Table

	payment_id	customer_id	payment_date	formatted_payment_date
1	P001	101	2026-01-05	2026-01-05
2	P002	102	2026-02-10	2026-02-10
3	P003	103	2026-03-15	2026-03-15
4	P004	104	2026-04-20	2026-04-20
5	P005	105	2026-05-25	2026-05-25

Question 9: Days Since Last Purchase

```sql
-- Q9: Days Since Last Purchase
CREATE OR REPLACE TABLE customer_purchases (
  customer_id        INT,
  customer_name      STRING,
  last_purchase_date DATE
);
INSERT INTO customer_purchases VALUES
  (501, 'John',  '2026-05-01'),
  (502, 'Mary',  '2026-05-10'),
  (503, 'Peter', '2026-05-15'),
  (504, 'Sarah', '2026-05-20'),
  (505, 'Thabo', '2026-05-25');

SELECT
  customer_id,
  customer_name,
  last_purchase_date,
  DATEDIFF(CURRENT_DATE(), last_purchase_date) AS days_since_last_purchase
FROM customer_purchases;
```

Add parameter

Results 3 of 3 | Table

	customer_id	customer_name	last_purchase_date	days_since_last_purchase
1	501	John	2026-05-01	92
2	502	Mary	2026-05-10	83
3	503	Peter	2026-05-15	78
4	504	Sarah	2026-05-20	73
5	505	Thabo	2026-05-25	68

Question 10: Expected Delivery Date

```sql
-- Q10: Expected Delivery Date
CREATE OR REPLACE TABLE shipping_orders (
  order_id    INT,
  customer_id INT,
  order_date  DATE
);
INSERT INTO shipping_orders VALUES
  (6001, 101, '2026-05-01'),
  (6002, 102, '2026-05-03'),
  (6003, 103, '2026-05-05'),
  (6004, 104, '2026-05-07'),
  (6005, 105, '2026-05-09');

SELECT
  order_id,
  customer_id,
  order_date,
  DATE_ADD(order_date, 7) AS expected_delivery_date
FROM shipping_orders;
```

Add parameter

Results 3 of 3 | Table

	order_id	customer_id	order_date	expected_delivery_date
1	6001	101	2026-05-01	2026-05-08
2	6002	102	2026-05-03	2026-05-10
3	6003	103	2026-05-05	2026-05-12
4	6004	104	2026-05-07	2026-05-14
5	6005	105	2026-05-09	2026-05-16

Question 11: Year, Month, Day Separately

```sql
-- Q11: Year, Month, Day Separately
CREATE OR REPLACE TABLE bookings (
  booking_id   STRING,
  customer_id  INT,
  booking_date DATE
);
INSERT INTO bookings VALUES
  ('B001', 101, '2026-01-12'),
  ('B002', 102, '2026-02-18'),
  ('B003', 103, '2026-03-22'),
  ('B004', 104, '2026-04-09'),
  ('B005', 105, '2026-05-27');

SELECT
  booking_id,
  customer_id,
  booking_date,
  YEAR(booking_date)  AS booking_year,
  MONTH(booking_date) AS booking_month,
  DAY(booking_date)   AS booking_day
FROM bookings;
```

Add parameter

Results 3 of 3 | Table

	booking_id	customer_id	booking_date	booking_year	booking_month	booking_day
1	B001	101	2026-01-12	2026	1	12
2	B002	102	2026-02-18	2026	2	18
3	B003	103	2026-03-22	2026	3	22
4	B004	104	2026-04-09	2026	4	9
5	B005	105	2026-05-27	2026	5	27

Question 12: Filter Orders from 2026

```sql
-- Q12: Filter Orders from 2026
CREATE OR REPLACE TABLE yearly_orders (
  order_id    INT,
  customer_id INT,
  order_date  DATE,
  amount      DECIMAL(10,2)
);
INSERT INTO yearly_orders VALUES
  (7001, 101, '2024-12-15', 500),
  (7002, 102, '2025-01-20', 1200),
  (7003, 103, '2025-06-10', 800),
  (7004, 104, '2026-02-05', 1500),
  (7005, 105, '2026-05-25', 2000);

SELECT
  order_id,
  customer_id,
  order_date,
  YEAR(order_date) AS order_year,
  amount
FROM yearly_orders
WHERE YEAR(order_date) = 2026;
```

Add parameter

Results 3 of 3 | Table

	order_id	customer_id	order_date	order_year	amount
1	7004	104	2026-02-05	2026	1500.00
2	7005	105	2026-05-25	2026	2000.00

Question 13: Filter Orders from March

```sql
-- Q13: Filter Orders from March
CREATE OR REPLACE TABLE monthly_orders (
  order_id    INT,
  customer_id INT,
  order_date  DATE,
  amount      DECIMAL(10,2)
);
INSERT INTO monthly_orders VALUES
  (8001, 101, '2026-01-15', 500),
  (8002, 102, '2026-02-20', 1200),
  (8003, 103, '2026-03-10', 800),
  (8004, 104, '2026-03-25', 1500),
  (8005, 105, '2026-05-30', 2000);

SELECT
  order_id,
  customer_id,
  order_date,
  MONTH(order_date) AS order_month,
  amount
FROM monthly_orders
WHERE MONTH(order_date) = 3;
```

Add parameter

Results 3 of 3 | Table

	order_id	customer_id	order_date	order_month	amount
1	8003	103	2026-03-10	3	800.00
2	8004	104	2026-03-25	3	1500.00

Question 14: Last Day of the Month

```
-- Q14: Last Day of the Month
CREATE OR REPLACE TABLE subscriptions (
  subscription_id STRING,
  customer_id     INT,
  start_date      DATE
);
INSERT INTO subscriptions VALUES
  ('SUB001', 101, '2026-01-10'),
  ('SUB002', 102, '2026-02-15'),
  ('SUB003', 103, '2026-03-20'),
  ('SUB004', 104, '2026-04-25'),
  ('SUB005', 105, '2026-05-30');

SELECT
  subscription_id,
  customer_id,
  start_date,
  LAST_DAY(start_date) AS month_end_date
FROM subscriptions;
```

Add parameter

Results 3 of 3 — Table

	order_id	customer_id	order_date	order_month	amount
1	8003	103	2026-03-10	3	800.00
2	8004	104	2026-03-25	3	1500.00

Question 15: First Day of the Month

```
-- Q15: First Day of the Month
CREATE OR REPLACE TABLE campaign_sends (
  send_id     STRING,
  customer_id INT,
  send_date   DATE
);
INSERT INTO campaign_sends VALUES
  ('C001', 101, '2026-01-12'),
  ('C002', 102, '2026-02-18'),
  ('C003', 103, '2026-03-22'),
  ('C004', 104, '2026-04-09'),
  ('C005', 105, '2026-05-27');

SELECT
  send_id,
  customer_id,
  send_date,
  DATE_TRUNC('MONTH', send_date) AS month_start_date
FROM campaign_sends;
```

Add parameter

Results 3 of 3 — Table

	send_id	customer_id	send_date	month_start_date
1	C001	101	2026-01-12	2026-01-01T00:00:00.000+00:...
2	C002	102	2026-02-18	2026-02-01T00:00:00.000+00:...
3	C003	103	2026-03-22	2026-03-01T00:00:00.000+00:...
4	C004	104	2026-04-09	2026-04-01T00:00:00.000+00:...
5	C005	105	2026-05-27	2026-05-01T00:00:00.000+00:...

Question 16: Format as Month and Year

```
-- Q16: Format as Month and Year
CREATE OR REPLACE TABLE invoice_dates (
  invoice_id   STRING,
  customer_id  INT,
  invoice_date DATE
);
INSERT INTO invoice_dates VALUES
  ('INV001', 101, '2026-01-05'),
  ('INV002', 102, '2026-02-10'),
  ('INV003', 103, '2026-03-15'),
  ('INV004', 104, '2026-04-20'),
  ('INV005', 105, '2026-05-25');

SELECT
  invoice_id,
  customer_id,
  invoice_date,
  DATE_FORMAT(invoice_date, 'MMMM yyyy') AS invoice_month_year
FROM invoice_dates;

-- Q17: Customer Age from Date of Birth
```

Add parameter

Results 3 of 3 | Table

	invoice_id	customer_id	invoice_date	invoice_month_year
1	INV001	101	2026-01-05	January 2026
2	INV002	102	2026-02-10	February 2026
3	INV003	103	2026-03-15	March 2026
4	INV004	104	2026-04-20	April 2026
5	INV005	105	2026-05-25	May 2026

Question 17: Customer Age from Date of Birth

```
-- Q17: Customer Age from Date of Birth
CREATE OR REPLACE TABLE customer_birthdays (
  customer_id   INT,
  customer_name STRING,
  date_of_birth DATE
);
INSERT INTO customer_birthdays VALUES
  (901, 'John',  '1998-05-10'),
  (902, 'Mary',  '1990-08-20'),
  (903, 'Peter', '2002-03-15'),
  (904, 'Sarah', '1985-12-01'),
  (905, 'Thabo', '2000-07-30');

SELECT
  customer_id,
  customer_name,
  date_of_birth,
  FLOOR(DATEDIFF(CURRENT_DATE(), date_of_birth) / 365.25) AS customer_age
FROM customer_birthdays;

-- Q18: Classify Weekend vs Weekday
```

Add parameter

Results 3 of 3 | Table

	customer_id	customer_name	date_of_birth	customer_age
1	901	John	1998-05-10	28
2	902	Mary	1990-08-20	35
3	903	Peter	2002-03-15	24
4	904	Sarah	1985-12-01	40
5	905	Thabo	2000-07-30	26

Question 18: Classify Weekend vs Weekday

```
366
367  -- Q18: Classify Weekend vs Weekday
368  CREATE OR REPLACE TABLE weekend_orders (
369    order_id    INT,
370    customer_id INT,
371    order_date  DATE
372  );
373  INSERT INTO weekend_orders VALUES
374    (9001, 101, '2026-05-01'),
375    (9002, 102, '2026-05-02'),
376    (9003, 103, '2026-05-03'),
377    (9004, 104, '2026-05-04'),
378    (9005, 105, '2026-05-05');
379
380  SELECT
381    order_id,
382    customer_id,
383    order_date,
384    DAYNAME(order_date) AS day_name,
385    CASE
386      WHEN DAYNAME(order_date) IN ('Saturday', 'Sunday') THEN 'Weekend'
387      ELSE 'Weekday'
388    END AS day_type
389  FROM weekend_orders;
```

Add parameter

Results 3 of 3 | Table

	order_id	customer_id	order_date	day_name	day_type
1	9001	101	2026-05-01	Fri	Weekday
2	9002	102	2026-05-02	Sat	Weekday
3	9003	103	2026-05-03	Sun	Weekday
4	9004	104	2026-05-04	Mon	Weekday
5	9005	105	2026-05-05	Tue	Weekday

Question 19: Transactions by Quarter

```
390
391  -- Q19: Transactions by Quarter
392  CREATE OR REPLACE TABLE quarterly_transactions (
393    transaction_id   STRING,
394    customer_id      INT,
395    transaction_date DATE,
396    amount           DECIMAL(10,2)
397  );
398  INSERT INTO quarterly_transactions VALUES
399    ('Q001', 101, '2026-01-15', 500),
400    ('Q002', 102, '2026-03-20', 1200),
401    ('Q003', 103, '2026-04-10', 800),
402    ('Q004', 104, '2026-07-05', 1500),
403    ('Q005', 105, '2026-10-25', 2000);
404
405  SELECT
406    transaction_id,
407    customer_id,
408    transaction_date,
409    QUARTER(transaction_date) AS transaction_quarter,
410    amount
411  FROM quarterly_transactions;
412
```

Add parameter

Results 3 of 3 | Table

	transaction_id	customer_id	transaction_date	transaction_quarter	amount
1	Q001	101	2026-01-15	1	500.00
2	Q002	102	2026-03-20	1	1200.00
3	Q003	103	2026-04-10	2	800.00
4	Q004	104	2026-07-05	3	1500.00
5	Q005	105	2026-10-25	4	2000.00

Question 20: Orders Older Than 30 Days

```
412
413  -- Q20: Orders Older Than 30 Days
414  CREATE OR REPLACE TABLE recent_orders (
415    order_id    STRING,
416    customer_id INT,
417    order_date  DATE,
418    amount      DECIMAL(10,2)
419  );
420  INSERT INTO recent_orders VALUES
421    ('R001', 101, '2026-04-01', 500),
422    ('R002', 102, '2026-04-15', 1200),
423    ('R003', 103, '2026-05-01', 800),
424    ('R004', 104, '2026-05-10', 1500),
425    ('R005', 105, '2026-05-25', 2000);
426
427  SELECT
428    order_id,
429    customer_id,
430    order_date,
431    DATEDIFF(CURRENT_DATE(), order_date) AS days_since_order,
432    amount
433  FROM recent_orders
434  WHERE DATEDIFF(CURRENT_DATE(), order_date) > 30;
```

Add parameter

Results 3 of 3 — Table

	order_id	customer_id	order_date	days_since_order	amount
1	R001	101	2026-04-01	122	500.00
2	R002	102	2026-04-15	108	1200.00
3	R003	103	2026-05-01	92	800.00
4	R004	104	2026-05-10	83	1500.00
5	R005	105	2026-05-25	68	2000.00

Bonus Question

```
441  CREATE OR REPLACE TABLE customer_recency (
442    customer_id       INT,
443    customer_name     STRING,
444    last_purchase_date DATE,
445    total_spend       DECIMAL(10,2)
446  );
447  INSERT INTO customer_recency VALUES
448    (1001, 'John', '2026-05-25', 5000),
449    (1002, 'Mary', '2026-05-10', 2500),
450    (1003, 'Peter', '2026-04-01', 700),
451    (1004, 'Sarah', '2026-02-15', 15000),
452    (1005, 'Thabo', '2025-12-20', 300);
453
454  SELECT
455    customer_id,
456    customer_name,
457    last_purchase_date,
458    DATEDIFF(CURRENT_DATE(), last_purchase_date) AS days_since_last_purchase,
459    CASE
460      WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date) <= 30 THEN 'Active Customer'
461      WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date) BETWEEN 31 AND 90 THEN 'At Risk Customer'
462      ELSE 'Inactive Customer'
463    END AS customer_status
464  FROM customer_recency;
```

Add parameter

Results 3 of 3 — Table

	customer_id	customer_name	last_purchase_date	days_since_last_purchase	customer_status
1	1001	John	2026-05-25	68	At Risk Customer
2	1002	Mary	2026-05-10	83	At Risk Customer
3	1003	Peter	2026-04-01	122	Inactive Customer
4	1004	Sarah	2026-02-15	167	Inactive Customer
5	1005	Thabo	2025-12-20	224	Inactive Customer